



MODULE NAME:	MODULE CODE:
PRINCIPLES OF SECURITY	PRSE6212/w

ASSESSMENT TYPE: ASSIGNMENT 2 (PAPER ONLY)

TOTAL MARK ALLOCATION: 100 MARKS

TOTAL HOURS: 10 HOURS

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. ***No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
2. ***Your assignment must be submitted through Turnitin.***
3. ***Save a copy of your assignment before submitting it.***
4. ***Assignments must be typed unless otherwise specified.***
5. ***All work must be adequately and correctly referenced.***
6. ***This is an individual assignment.***

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where both minor and major errors** (e.g. two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g., alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using introductory phrases, accurate grammar, etc.	Minor incongruities: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g. publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. - An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	Major incongruities: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

Question 1:**(Marks: 15)**

In his argument, Ciampa (2024) states that *"The relationship between security and convenience is not directly proportional."* Using a practical example from the current South African banking sector, defend this argument. Your response should include the following:

1. **Introduction (3 marks):**

Briefly explain Ciampa's argument and provide an overview of the inverse relationship between security and convenience in the context of digital banking.

2. **Practical Example (9 marks):**

Provide a current example from the South African banking sector that illustrates how increased security measures have reduced convenience. Discuss how security features, such as multi-factor authentication (MFA), biometric authentication, or mobile banking app security protocols, have impacted user convenience. Support your argument with facts or cases from South African banks.

3. **Present a brief conclusion (3 marks).**

Instructions:

In your response, ensure you:

- Use clear and relevant examples.
- Use the example to highlight the inverse proportionality between security and convenience.

Question 2**(Marks: 10)****Context:**

Biometric authentication, despite its potential for secure identification, faces several challenges that make its adoption in environments like South African government secondary schools difficult.

Question:

In your response, address the following:

• **Security Challenges and Concerns (6 marks):**

Discuss why South African government secondary schools might be hesitant to implement biometric authentication systems for pupils, focusing on security-related challenges. Consider issues such as false acceptance rates (FAR), false rejection rates (FRR), and the risk of biometric systems being "tricked." How do these concerns impact the decision to use biometrics in schools?

- **Practical examples (4 marks):**

Provide examples of possible biometric authentication methods that could be implemented in South African secondary schools. Discuss how these methods might work in practice and the potential security risks associated with each.

Instructions:

- Support your response with relevant examples of security concerns related to biometric authentication.
- Consider the balance between the security benefits and the risks of fraud or system errors.

Question 3

(Marks: 25)

Context:

Understanding the difference between DNS poisoning and DNS hijacking is crucial in network security.

Question:

In your response, please address the following sections:

1. **Definitions and Differences (12 marks):**

Define DNS poisoning and DNS hijacking. Clearly explain how these two types of attacks differ in terms of their methods, goals, and impact on the DNS (Domain Name System).

2. **Intended Harm (8 marks):**

Briefly discuss the intended harm caused by each type of attack. What are the consequences for users and organizations that fall victim to these attacks?

3. **Effectiveness of DNS Poisoning (5 marks):**

Investigate whether DNS poisoning attacks are still effective in the current cybersecurity landscape, especially in light of widespread HTTPS adoption. What implications does HTTPS have for the effectiveness of DNS poisoning?

Instructions:

- Your answers should be concise yet comprehensive, using examples where necessary.
- Provide evidence in the form of references for any real-world incidents, case studies, or technical information you cite in your answers.

Question 4**(Marks: 25)****Question:**

Briefly discuss the role of Public Key Infrastructures (PKI) in cryptography, focusing on its importance within the South African banking sector.

Additionally, analyse the role of PKI Trust Models—Hierarchical, Distributed, and Bridge Trust models—in enhancing security and trust in this context.

Instructions:

- Your response should be well-structured and concise, demonstrating a clear understanding of PKI and its trust models.
- Use relevant examples from the South African banking industry to support your discussion.
- Provide evidence in the form of references for any case studies or technical information you cite in your answer.

Question 5**(Marks: 10)****Question:**

Discuss the additional wireless security protections that South African Small, Medium, and Micro Enterprises (SMMEs) must consider when installing and configuring a wireless LAN (WLAN). In your response, address the following points:

1. **Optimal Installation Practices:**
Explain the importance of wireless access point (AP) placement and signal strength settings in ensuring both coverage and security.
2. **Configuration Options:**
Discuss specific configuration options available for APs that enhance security, including spectrum selection and antenna placement.

Instructions:

- Your response should be concise yet comprehensive, demonstrating a clear understanding of wireless security protections for SMMEs.
- Provide examples where relevant and reference any sources used in your answer.

Question 6**(Marks: 15)****Question:**

Write a letter addressed to the citizens of South Africa that explains the benefits of electricity smart meters compared to traditional analogue meters. In your letter, please:

1. **Investigate Resistance:**

Briefly discuss why some South Africans are resisting the adoption of smart meters.

2. **Address Concerns:**

Use the article "Unlocking the future: Smart metering redefines utility management and sustainability" by Francois Conradie (available at:

<https://infrastructurenews.co.za/2024/10/14/unlocking-the-future-smart-metering-redefines-utility-management-and-sustainability/>) and other relevant sources to address the concerns raised by citizens.

3. **Benefits of Smart Meters:**

Highlight the key advantages of smart meters over analogue meters, emphasizing their impact on utility management and sustainability.

Instructions:

- Ensure your article is well-structured and engaging for the readers.
- Provide in-text references for any material used from the article by Francois and other sources.
- Your response should demonstrate a clear understanding of smart meters and their benefits to the South African context.

[TOTAL MARKS: 100]